Audencia Business School

MSc Full Time

Master Thesis — Literature Review & Initial Methodology

Title:

"How Does the Level of Transparency in AI-Generated Recommendations Influence Consumer Trust and Willingness to Pay?"

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Academic Year 2025–2026

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Abstract

This study investigates how the level of transparency in AI-generated product recommendations influences consumer trust and willingness to pay (WTP) in e-commerce contexts. Despite the widespread deployment of AI recommendation systems across digital marketplaces, the causal relationship between algorithmic explainability, trust formation, and monetary consumer valuation remains empirically underexplored. Drawing on the Explainable AI (XAI) literature, algorithm aversion and appreciation theory, and the trust–WTP pathway established in prior e-commerce research, this study tests whether providing explanations for AI-generated recommendations increases consumer trust and, through trust, consumers' willingness to pay a price premium for recommended products.

A quantitative experimental design is employed. Participants are randomly assigned to one of three conditions: a human expert recommendation, an opaque AI recommendation, or a transparent AI recommendation. Trust is measured using validated 7-point Likert scales, and willingness to pay is estimated through conjoint analysis with four attributes — price, recommendation source, explanation level, and brand familiarity — capturing revealed preferences from forced trade-off decisions. The target sample comprises 300–400 adults aged 18–45 who actively shop online.

The study addresses four gaps in the existing literature: the absence of experimental causal evidence in e-commerce recommendation contexts; the reliance on self-reported purchase intention rather than monetary WTP as an outcome variable; the limited generalisability of XAI findings from high-stakes professional domains to consumer purchasing; and the lack of economic quantification of transparency's value. The combined experimental and conjoint methodology produces an original empirical contribution by establishing a causal, monetarily quantified link between AI transparency, trust, and willingness to pay.

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1. Introduction

Artificial intelligence has become a defining feature of the modern digital marketplace. Recommendation systems powered by machine learning algorithms now guide consumer choices across the world's largest e-commerce platforms, accounting for a substantial share of platform revenues and individual purchase decisions. Yet, as these systems have grown more sophisticated, consumer concerns about their opacity, fairness, and privacy implications have intensified. Central to these concerns is the concept of transparency: whether consumers are informed about why an AI system is making a particular recommendation.

The core research question of this study is: How does the level of transparency in AI-generated recommendations influence consumer trust and willingness to pay? This question is motivated by a consistent gap in the existing literature. While prior research has established that AI personalisation can build trust and that trust drives purchase intention, the causal chain from explanation to trust to economic behaviour has not been tested experimentally in an e-commerce product recommendation context. Furthermore, most studies use self-reported purchase intention as their outcome measure; almost none translate trust into a monetary willingness to pay estimate.

Key terms are defined as follows. AI transparency refers to the degree to which an AI system provides intelligible, outcome-level explanations of its recommendations to end users. Consumer trust refers to the cognitive and affective confidence a consumer places in an AI recommendation system, encompassing both competence trust — the belief that the system works accurately — and goodwill trust — the belief that the system acts in the consumer's interest. Willingness to pay (WTP) refers to the maximum monetary amount a consumer would accept to pay for a product or feature, used here as a behavioural measure of the economic value attached to AI-driven recommendations.

The theoretical framework draws on three bodies of literature: the Explainable AI (XAI) literature, which theorises how algorithmic explanations affect human decision-making (Shulner-Tal et al., 2024; Von Zahn et al., 2025); the algorithm aversion and appreciation literature, which documents the behavioural paradoxes consumers exhibit when interacting with algorithmic versus human advisors (Kirshner, 2025; Gupta et al., 2026; Talebi et al., 2026); and the consumer trust literature in e-commerce, which establishes trust as the central mediating mechanism between AI input and purchase behaviour (Singhal et al., 2025; Mani et al., 2025). The study employs a between-subjects experimental design with three conditions integrated with conjoint analysis to produce attribute-level WTP estimates. This document presents the literature review (Section 2) and the initial research methodology (Section 3).

2. Literature Review

Digital commerce has become a vital part of the global economy, reshaping how consumers shop and how companies conduct business. As the volume and speed of e-commerce transactions grow, businesses have turned to intelligent systems capable of processing vast amounts of information rapidly. Artificial intelligence now powers recommendation engines, chatbots, dynamic pricing, and predictive analytics, enabling personalised experiences at scale. Yet persistent challenges — algorithmic bias, privacy concerns, and the opaque "black box" nature of many AI systems — raise critical questions about transparency, trust, and consumer welfare (Lim & Kim, 2025; Hassan et al., 2025). The following review synthesises the existing literature across six interrelated areas, building toward a theoretically grounded research design and a clear statement of the research gap.

2.1 AI Recommendations in Digital Markets: Personalisation, Trust, and Its Fragility

AI-powered recommendation systems are information-filtering tools designed to match users with products based on behavioural history and preferences. Their commercial stakes are substantial, but the conditions under which they succeed — or backfire — have become a central concern in recent scholarship.

The foundational insight from recent quantitative studies is that AI personalisation raises conversion and purchase intention primarily through trust as a mediator. Singhal et al. (2025), using structural equation modelling (SEM) with data from 300 online shoppers, found that AI-driven personalisation is a significant positive predictor of consumer trust ($\beta = 0.52$) and, through trust, of purchase intention ($\beta = 0.46$ indirect). The study also confirmed that privacy concerns negatively moderate the personalisation–trust relationship ($\beta = -0.18$), and the authors explicitly called for Explainable AI as a practical solution to privacy-eroded trust without testing it experimentally — a gap the present study directly addresses. Mani et al. (2025) replicated this at larger scale with 600 online consumers, demonstrating that AI system quality improves both consumer trust (cognitive) and user experience (affective), together explaining 55% of AI's total influence on online buying behaviour, with a 42% increase in conversion likelihood. Their findings also showed that privacy concerns ($\beta = -0.31$) and over-automation ($\beta = -0.27$) act as trust-eroding counterweights, illustrating what they term a "trust fragility" effect under conditions of opacity.

Jith et al. (2025), in a mixed-methods study combining survey data from 847 respondents with qualitative interviews from 32 participants, confirmed that AI personalisation strongly predicts purchase intention ($\beta = 0.68$) and that transparency ($\beta = 0.54$) and perceived control ($\beta = 0.47$)

serve as the primary mediators of the personalisation–trust relationship. A critical finding from their work is that the positive effect of personalisation on purchase intention dropped by 29% as privacy concern levels increased from low to high — quantifying the practical significance of consumer concerns about opacity.

A theoretically important counterpoint runs through this literature: personalisation alone is insufficient, and under certain conditions it actively undermines trust. Hassan et al. (2025), analysing 729 e-commerce consumers through SEM, found a negative moderation coefficient of −0.085, showing that poorly calibrated or unexplained personalisation can weaken the trust–satisfaction relationship. They attribute this to a "black box" effect: consumers who cannot understand why a recommendation was made cannot evaluate whether the system is working in their interest. This finding establishes that the transparency and explainability of AI recommendations — not merely their personalisation depth — critically shape commercial effectiveness.

Gao and Liang (2025), studying AI-powered virtual try-on technology among Chinese university students (n = 366), demonstrated through PLS-SEM that brand trust significantly moderates the pathways from AI-perceived value to impulsive buying intention, strengthening all three value dimensions — utilitarian, hedonic, and immersive. Their study found that brand trust reduces consumer scepticism toward AI-powered technology and streamlines the decision-making process. However, the study did not test whether transparency in AI recommendations itself generates that trust, and it focused exclusively on impulse buying in a try-on context, limiting its generalisability to deliberate product recommendation settings. The present study addresses both gaps.

2.2 Transparency and Explainability in AI (XAI): From Theory to Consumer Context

The concept of transparency in AI systems has been well theorised in computer science but inconsistently applied in consumer-facing contexts. The XAI literature distinguishes between model-level explanations — describing how a system works overall — and outcome-level explanations — describing why a specific decision was made. For consumers receiving product recommendations, the relevant form is almost always outcome-level: why did the system recommend this to me?

The most directly relevant empirical precedent is Peters et al. (2020), who conducted an online lab experiment using a Name-Your-Own-Price (NYOP) bidding method with 195 German participants to measure consumers' actual willingness to pay for transparency features of an AI credit-scoring system. Using SEM grounded in the Theory of Planned Behavior, they found a median WTP of €20 for transparency features, with trust as the central mediating pathway. Critically, they called for their findings to be examined in other contexts, including e-commerce — a call the present study answer directly. Furthermore, Peters et al. did not include a human expert comparison condition, used a single-package WTP measure that could not isolate the value of individual transparency attributes, and acknowledged single-method bias risk. The present study addresses all three limitations through its three-group experimental design and attribute-level conjoint analysis.

Shulner-Tal et al. (2024) extended this question into a large-scale between-subjects experiment with 3,546 participants on how explanation styles affect perceptions of AI versus human decision-making in a hiring simulation. They found that AI decision-making was consistently perceived as less fair than human decision-making when no explanation was provided, but that specific XAI approaches — particularly certification-based and sensitivity-based explanations — could narrow or close this fairness gap entirely. This finding has direct implications for e-commerce: how transparency is communicated, not merely whether it is present, determines consumer response. Importantly, Shulner-Tal et al. (2024) noted as an explicit limitation that prior AI perceptions were not measured, leaving uncontrolled variation in AI attitudes as a potential confound — a gap the present study addresses by measuring prior AI attitude as both a covariate and moderator.

Von Zahn et al. (2025) contributed a complementary cognitive mechanism, showing that XAI improves human metacognition through two professional experiments involving real estate agents (n = 149) and finance professionals (n = 200). Specifically, XAI led to more calibrated delegation decisions — users became more accurate in judging their own competence and when to defer to algorithmic advice. For e-commerce, this implies that transparent recommendations may help consumers make better-calibrated decisions about when to follow or override AI suggestions, supporting both trust and purchase conversion. The study's limitation is that it was conducted in a static environment without performance feedback and used global explanations (SHAP plots) rather than the outcome-level explanations most relevant to consumer recommendation contexts.

Balakrishnan et al. (2026) provided a related quantitative finding in a study of 359 participants: people exhibit systematic naïve advice-weighting behaviour that worsens prediction accuracy by 20–61%, but feature transparency reduces this miscalibration by 25%. This establishes that transparency does not merely improve subjective trust but produces measurable improvements in decision quality — a finding that supports the economic significance of transparency examined in the present study.

Lim and Kim (2025) demonstrated a "backfire" effect in the hospitality sector: AI-powered personalisation negatively affected consumer trust and platform usage when ethical AI framing was absent, but when ethical AI guidelines and empowerment features were present, perceived discrimination dropped and consumer responses improved. Their regression analyses showed that AI-powered personalisation had positive and significant impacts on perceived discrimination and privacy concerns in the absence of ethical framing. Critically, their study tested ethical AI only in a binary way — present versus absent — without examining what type or degree of transparency framing matters and used general psychological appraisals as mediators rather than AI-specific mechanisms such as algorithmic explainability. The present study's three-group design addresses this gap directly.

Taken together, the XAI literature establishes that transparent AI changes the quality and calibration of human decision-making. A consistent limitation is the distance of these studies from the consumer e-commerce context: they are conducted in credit scoring, hiring, real estate, or financial services. The present study specifically tests XAI transparency in a consumer product recommendation setting.

2.3 Algorithm Aversion vs. Algorithm Appreciation: Behavioural Paradoxes

A central paradox in consumer-AI interaction research is that people do not respond uniformly to algorithmic advice. Two competing patterns have been documented — algorithm aversion, the distrust or rejection of algorithmic recommendations even when objectively superior, and algorithm appreciation, the systematic preference for algorithmic over human advice — and the interplay between them forms the behavioural tension the present study's hypotheses are designed to interrogate.

Kirshner (2025) integrated Construal Level Theory (CLT) to argue that people perceive algorithms as psychologically more distant than human advisors, which explains why algorithm aversion decreases for temporally or spatially distant tasks. A key finding was that while people perceive algorithmic agents' outputs at low construal levels, they simultaneously perceive these same agents at a far psychological distance, creating a tension between familiarity and abstraction. For e-commerce, this implies that consumers may be more algorithmically receptive when purchasing for future use or unfamiliar product categories. Importantly, Kirshner's between-subjects studies produced null effects, suggesting that the distance-aversion relationship is stronger in within-subjects designs and may be sensitive to how consumers are asked to evaluate their options — a methodological consideration for the present study's between-subjects design.

Gupta et al. (2026), using Fuzzy Analytic Hierarchy Process (F-AHP) with 184 respondents drawn from AI experts in academia and industry, found that personality factors — particularly self-esteem, self-efficacy, neuroticism, and locus of control — are the strongest predictors of algorithm aversion, stronger than task type or domain familiarity. Their study also established that subjective and moral tasks increase algorithm aversion while objective and complex tasks increase algorithm use. This personality-driven account supports the present study's inclusion of prior AI attitude as a moderating variable. A limitation acknowledged by the authors is the expert-only sample and Indian context, which may not reflect general consumer populations.

Talebi et al. (2026) introduced a novel situational antecedent to algorithm appreciation: marketplace discrimination. Across three laboratory studies and one field study, they found that consumers who experienced discrimination from human service providers showed significantly higher preference for AI recommendations as a coping mechanism, mediated by perceived embarrassment. This extends UTAUT by positioning AI adoption as an emotionally driven response to interpersonal experience. For the present study, it underscores that the human expert baseline condition is not a neutral default but carries social valence that may actively disadvantage it with certain consumer segments. The paper's limitation is that AI preference was stronger in public consumption settings where social judgment is salient, suggesting a boundary condition that may not apply uniformly in online private purchasing environments.

Al Helaly et al. (2023) revealed a transparency paradox in online advertising: even when consumers display defensive psychological responses — scepticism, confusion, and avoidance — their actual engagement remained paradoxically high, especially when personalisation benefits were perceived. Conducted through two quantitative Qualtrics experiments manipulating ad personalisation and transparency, their study established that the Defensive Response Model explains why non-transparent AI recommendations may produce attitudinal resistance while failing to reduce behavioural engagement. A key limitation is that the study measured engagement clicks rather than purchase intention or WTP, and trust was never modelled as a mediating mechanism. The present study fills this gap by placing trust as the construct bridging transparency and economic behaviour.

A key limitation of the aversion/appreciation literature is its almost exclusive focus on preference or attitude as the outcome variable. Studies rarely proceed to measure the monetary implications of these preferences — how much more, or less, would an algorithm-appreciating versus algorithm-averse consumer pay for a transparent recommendation? The present study's conjoint methodology bridges this gap directly.

2.4 Consumer Trust in AI: Dimensions, Antecedents, and Mediation Pathways

Trust is the central psychological mechanism through which transparency is theorised to affect willingness to pay. Consumer trust in AI encompasses both cognitive components — rational assessment of system competence and reliability — and affective components — emotional confidence and goodwill. This distinction matters because, as Lu et al. (2025) demonstrated using data from 494 Chinese online shoppers and SEM, competence trust and goodwill trust respond differently to platform behaviour: competence trust leads consumers to tolerate questionable algorithmic behaviour through moral decoupling, while goodwill trust makes consumers more sensitive to perceived unfairness. Specifically, moral decoupling was found to mediate the relationship between algorithmic price discrimination and purchase intention, and these two trust dimensions moderated this mediation differently. These two dimensions likely respond differently to transparency manipulations — a nuance the present study's experimental design is positioned to explore.

Singhal et al. (2025) provided foundational SEM evidence that trust fully mediates the personalisation–purchase intention relationship and explicitly called for Explainable AI as a practical solution to privacy-eroded trust without testing it. Yu et al. (2025), modelling consumer reactions to AI-generated content through a trust–risk dual pathway framework with 507 e-commerce users, found that AI content quality raises purchase intention by simultaneously increasing trust and reducing perceived risk, with platform responsibility amplifying both pathways. Their study used a two-phase mixed-method approach combining systematic literature review, expert interviews, and SEM, establishing that perceived platform responsibility acts as a "dual amplification" mechanism — strengthening positive trust effects while also making users more sensitive to risks. This finding supports positioning AI transparency as a form of responsible platform behaviour that reduces perceived risk and supports conversion.

Moliner-Tena and Tortosa-Edo (2025), using data from 1,123 Spanish consumers analysed through SEM and dual privacy calculus theory, confirmed that online customer experience (OCX) is the strongest antecedent of privacy concerns, operating through both trust and satisfaction mechanisms. E-trust reduces privacy concerns through the risk calculus mechanism, while e-satisfaction does so through the privacy calculus mechanism. This supports the positioning of AI transparency as a dimension of customer experience that, when positive, reduces privacy-driven trust erosion. A limitation is that the study did not examine AI recommendations or transparency as discrete variables, relied on observational survey data, and found that trust mechanisms differ between online-only and omnichannel consumer journeys in ways that complicate generalisation.

Shin (2025), applying Privacy Calculus Theory to 336 Korean digital news consumers, found that perceived control and risk significantly affect users' willingness to disclose personal data, and that privacy-sensitive users are actually more willing to pay for transparent, personalised experiences — suggesting that transparency can shift economic behaviour even among those most concerned about data use. This finding provides direct behavioural grounding for the present study's hypothesis that AI transparency can meaningfully shift WTP. A limitation is the single-country sample, skew toward urban older adults, and media consumption context, which differs meaningfully from active e-commerce purchase decisions.

Moayery and Urbonavičius (2025), studying the privacy paradox across two studies (216 Lithuanian students and 498 US adults), found that stronger online shopping habits weaken the link between privacy intentions and actual disclosure behaviour, and that prevention-focused individuals follow their intentions more consistently than promotion-focused individuals. This finding is relevant to WTP measurement: self-reported WTP may diverge from actual behaviour, which the present study's conjoint methodology partially addresses by measuring revealed preferences through forced trade-off decisions rather than direct self-report.

Saha (2025), in a conceptual theoretical framework, proposed that consumer trust acts as the central mediating variable between AI-generated advertising and purchase intention, and that transparency positively moderates this relationship. The proposed influence pathway moves from visual stimulation through ethical evaluation to cognitive trust to behavioural intent. Critically, however, this study is entirely conceptual — no primary data were collected, no experimental conditions were tested, no statistical findings are reported, and no WTP estimates are provided. The present study addresses all four of these limitations.

2.5 Willingness to Pay: The Missing Economic Endpoint

Willingness to pay (WTP) is the terminal outcome variable in this study — the point at which psychological trust is translated into economic value. While the trust literature is well-developed, the connection between AI-specific transparency and monetary WTP has received surprisingly limited empirical attention.

Peters et al. (2020) remains the most direct precedent. Their NYOP bidding experiment demonstrated a median WTP of €20 for transparency features in AI systems, with trust as the key mediating variable. However, their domain was credit scoring, their WTP measure priced transparency as a single bundled feature, and they explicitly called for replication in other contexts. The present study extends their work to an e-commerce product recommendation context and uses conjoint analysis to decompose WTP at the attribute level — producing a more granular and actionable estimate.

Conjoint analysis is particularly well-suited to WTP estimation because it measures revealed preferences through forced trade-off decisions rather than direct self-report. As Moayery and Urbonavičius (2025) demonstrated, people's stated privacy intentions often diverge from their actual disclosure behaviour — a gap that direct self-report WTP questions would replicate. Respondents in a conjoint exercise who choose a transparent AI recommendation over a cheaper opaque one reveal their marginal WTP for transparency through their choices, which is more behaviourally valid. Gao and Liang (2025) showed that brand trust significantly strengthens the pathway from AI-perceived value to impulse buying intention — directionally consistent with the claim that trust translates into economic premium. Lu et al. (2025) provided further evidence that trust dimensions shape purchase intention under conditions of perceived algorithmic unfairness, reinforcing trust as a necessary mediating variable in the transparency–WTP chain.

From the algorithm aversion/appreciation perspective, the WTP gap is particularly stark: the entire literature documents preferences and attitudes but stops before measuring their monetary value. Even the most quantitatively rigorous studies — Gupta et al. (2026) with F-AHP, Kirshner (2025) with CLT experiments, Talebi et al. (2026) with field experiments — establish that consumers prefer or reject algorithmic advice under various conditions without ever asking how much they would actually pay for the algorithmic option they prefer. This is the precise contribution of the present study's conjoint component: converting psychological preferences into economic estimates.

2.6 Algorithm Attitude as a Moderator

This study examines whether consumers' prior attitude toward AI moderates the effect of transparency on trust. Gupta et al. (2026) established that personality traits — particularly self-efficacy and locus of control — are the strongest individual-level predictors of algorithm aversion. Transparency, in this account, may function as a form of control restoration: by explaining its reasoning, the AI system reduces the perceived threat to consumer autonomy that black-box algorithms impose on sceptical consumers.

Conversely, consumers who already favour algorithmic advice may exhibit ceiling effects: if they already trust AI strongly, added transparency may produce diminishing marginal returns. This asymmetry is the theoretical basis for the moderation hypothesis: the effect of transparency on trust should be stronger for algorithm-sceptical consumers than for algorithm-appreciative ones, because the former has more room to gain. Shulner-Tal et al. (2024) noted as a limitation that prior AI perceptions were not measured in their study, leaving uncontrolled variation in AI attitudes as a potential explanation for between-subjects heterogeneity. Talebi et al. (2026) further showed that emotional experiences can shift algorithm attitude situationally, reinforcing that prior attitude is a dynamic moderator whose baseline level matters for understanding transparency effects. The present study measures prior AI attitude at the start of the survey and uses it as both a covariate in regression analysis and a moderator in interaction models.

2.7 Research Gap and Positioning of the Present Study

The literature reviewed above converges on four consistent limitations that the present study is specifically designed to address.

First, causal evidence is absent. Most studies examining AI transparency and trust use cross-sectional survey designs, which establish correlation but cannot support causal inference (Singhal et al., 2025; Jith et al., 2025; Hassan et al., 2025; Yu et al., 2025; Moliner-Tena & Tortosa-Edo, 2025; Mani et al., 2025). Peters et al. (2020) and Shulner-Tal et al. (2024) are notable experimental exceptions — but in financial and HR hiring contexts far removed from consumer e-commerce. The present study applies experimental logic specifically to e-commerce product recommendations, which is the domain where AI recommendations are most commercially consequential and most consumer-facing.

Second, the outcome variable is misaligned with economic reality. Prior research almost uniformly uses self-reported purchase intention as the outcome variable, a measure susceptible to social desirability bias that does not translate into a monetary estimate of how much transparency is worth (Singhal et al., 2025; Jith et al., 2025; Saha, 2025; Gao & Liang, 2025). The present study addresses both limitations by combining experimental trust measurement with conjoint-based WTP estimation, producing an attribute-level monetary estimate not provided by any existing study in the reviewed literature. Peters et al. (2020) measured WTP but as a single bundled feature in a non-e-commerce context; the present study goes further by decomposing WTP across recommendation source, explanation level, price, and brand familiarity.

Third, the domain is misspecified for consumer generalisation. The XAI literature focuses predominantly on high-stakes decision domains — credit, hiring, finance, demand forecasting (Peters et al., 2020; Shulner-Tal et al., 2024; Von Zahn et al., 2025; Balakrishnan et al., 2026) — where the consequences of AI error are severe and professional expertise is involved. Whether transparency effects operate similarly in lower-stakes, hedonic, or habitual consumer purchasing remains unexplored. The present study, targeting general online shoppers choosing consumer products, tests whether transparency effects generalise beyond expert-driven, high-stakes contexts.

Fourth, the aversion/appreciation literature is rich on preference but silent on economic value. Even when algorithm appreciators are shown to prefer AI (Kirshner, 2025; Gupta et al., 2026; Talebi et al., 2026; Al Helaly et al., 2023), we do not know how much more they will pay for a recommendation they trust. The present study's conjoint component produces exactly this estimate.

No existing study in the reviewed literature combines a three-group experimental design — human expert, opaque AI, transparent AI — with conjoint analysis in a single study. The two-method approach — the experimental component testing the transparency → trust chain, and the conjoint component quantifying the trust → WTP translation — together produce a complete empirical test of the chain from algorithmic explanation to monetary consumer value. The original academic contribution is therefore to add new, causally grounded and monetarily quantified knowledge to the XAI and consumer trust literature by establishing the transparency–trust–WTP chain in an e-commerce context.

3. Methodology

3.1 Research Approach and Justification

A quantitative experimental approach is employed to examine the causal relationship between AI transparency, consumer trust, and willingness to pay (WTP). This approach is chosen for two reasons. First, experimental designs allow for causal inference by controlling extraneous variables through random assignment — a methodological requirement given that most prior studies establish only correlational evidence (Singhal et al., 2025; Hassan et al., 2025; Jith et al., 2025). Cross-sectional survey studies like those of Mani et al. (2025) and Moliner-Tena and Tortosa-Edo (2025) confirm associations between AI personalisation, trust, and purchase intention but cannot establish the direction or magnitude of the causal effect. Second, conjoint analysis provides a behaviourally valid measure of WTP by capturing revealed preferences through forced trade-off decisions, rather than relying on direct self-report measures that are vulnerable to the intention–behaviour gap documented by Moayery and Urbonavičius (2025).

The study employs a between-subjects causal-comparative design. Participants are randomly assigned to one of three conditions: (A) a human expert recommendation (baseline), (B) an opaque AI recommendation with no explanation, or (C) a transparent AI recommendation that explains why the product was recommended. This three-group structure tests two distinct comparisons: condition B versus A isolates the effect of AI attribution on trust relative to a human expert; condition C versus B isolates the specific effect of transparency on trust and WTP while holding AI attribution constant. This design directly responds to Peters et al.'s (2020) limitation that no human expert comparison was included in their study.

The theoretical structure guiding the experimental design is the chain: AI Transparency → Consumer Trust → Willingness to Pay, with Prior AI Attitude as a moderating variable. This chain is grounded in the trust–purchase intention literature (Singhal et al., 2025), the metacognitive XAI framework (Von Zahn et al., 2025), the trust–risk dual pathway framework (Yu et al., 2025), and the conceptual trust-mediation model proposed by Saha (2025), all of which converge on trust as the central mediating mechanism between algorithmic explainability and consumer behaviour.

3.2 Experimental Conditions and Stimuli

Each participant will be exposed to a realistic product recommendation scenario on a mock e-commerce interface. The three experimental conditions differ only in the source and explanation framing of the recommendation:

Condition A — Human Expert Recommendation: The recommendation is attributed to a human product specialist with relevant expertise. No algorithmic element is mentioned. This condition serves as the social baseline, capturing the social valence of human judgment as documented by Talebi et al. (2026) and Shulner-Tal et al. (2024).

Condition B — Opaque AI Recommendation: The recommendation is attributed to an AI system, but no explanation is provided. This condition represents the standard "black box" deployment common across e-commerce platforms today and tests the baseline consumer response to unexplained AI, as studied by Hassan et al. (2025) and Lim & Kim (2025).

Condition C — Transparent AI Recommendation: The recommendation is attributed to an AI system, and an outcome-level explanation is provided (e.g., "We are recommending this because you have previously purchased similar items, rated comparable products highly, and users with your profile find this valuable"). This condition tests whether outcome-level explainability — the type found most effective by Shulner-Tal et al. (2024) — shifts trust and WTP relative to Condition B.

The rationale for including a human expert condition extends beyond providing a simple baseline. As reinforced by Talebi et al. (2026), AI recommendations carry different social valence than human recommendations — not necessarily inferior or superior but differently weighted depending on the consumer's prior AI attitude and emotional context. Condition A allows the present study to test not only whether transparent AI outperforms opaque AI but also whether transparent AI can match or exceed the trust generated by a human expert — a theoretically and practically important question for businesses deciding whether and how to deploy AI recommendation systems.

3.3 Conjoint Analysis Design

Conjoint analysis will be used to estimate attribute-level WTP for AI transparency. A choice-based conjoint (CBC) design is employed, in which participants are presented with a series of choice tasks, each offering three product recommendation options from which they must select one. Each option is described by a combination of the following four attributes and levels:

Attribute 1 — Price: €20 / €30 / €40 / €50. Price serves as the monetary scale against which the value of other attributes is measured. WTP for transparency is estimated as the monetary premium a participant accepts — revealed by their actual choices — for a transparent recommendation over an opaque one at the same or higher price.

Attribute 2 — Recommendation Source: Human expert / Opaque AI (no explanation) / Transparent AI (with explanation). This attribute directly operationalises the three experimental conditions and connects the conjoint task to the between-subjects manipulation.

Attribute 3 — Explanation Level: No explanation provided / Brief explanation (one sentence, e.g., "matches your previous purchases") / Detailed explanation (e.g., based on purchase history, ratings, and profile similarity). This attribute decomposes the transparency construct into two levels beyond the no-explanation baseline, allowing the study to test whether the depth of explanation matters beyond its mere presence — a distinction left unexplored by Lim & Kim (2025).

Attribute 4 — Brand Familiarity: Familiar brand / Unfamiliar brand. This attribute is included as a control to prevent brand recognition from confounding the attribution of WTP to the transparency manipulation, following Gao and Liang (2025), who showed brand trust moderates the AI-value–purchase pathway.

The choice tasks will be constructed using an orthogonal fractional factorial design to ensure efficient estimation of main effects and selected interactions while keeping respondent burden manageable. A minimum of 12 choice tasks per respondent is planned, consistent with best practice for CBC designs (Rao, 2014). Part-worth utility scores will be estimated for each attribute level, and marginal WTP for transparency will be calculated as the ratio of the transparency part-worth to the price sensitivity coefficient. This produces a monetary estimate — the additional euros a participant would pay for a transparent AI recommendation relative to an opaque one — extending Peters et al.'s (2020) NYOP approach by decomposing WTP at the attribute level rather than pricing transparency as a single bundled feature.

3.4 Data Collection Method

Primary data will be collected through an integrated online survey experiment administered via Qualtrics, targeting adults aged 18 to 45 who actively engage in online shopping. Qualtrics is chosen for its built-in random assignment functionality and its capacity to host conjoint tasks alongside Likert-scale measures. Respondents will be randomly assigned to one of the three recommendation scenarios to maintain experimental control.

After exposure to the assigned scenario, participants will complete a two-part assessment. Part 1 consists of validated 7-point Likert scales measuring consumer trust, adapted from the multi-dimensional trust scale distinguishing between competence trust and goodwill trust following the framework established by Lu et al. (2025). Part 2 consists of the 12-task conjoint choice exercise described in Section 3.3. The survey also collects individual-level variables including prior AI attitude (algorithm appreciation/aversion, adapted from Kirshner, 2025 and Gupta et al., 2026), privacy concerns (adapted from Singhal et al., 2025 and Moliner-Tena & Tortosa-Edo, 2025), and basic demographics. Prior AI attitude serves as both a covariate and a moderating variable in the regression analysis, following the theoretical rationale established in Section 2.6.

3.5 Sampling Method

The study targets adults aged 18 to 45 who actively shop online. A non-probability convenience sampling method is used, drawing participants from university networks, social media platforms, and online shopping communities, with a target sample size of 300–400 respondents. This sample size provides adequate statistical power for one-way ANOVA and regression analysis (Creswell & Creswell, 2018) and aligns with sample sizes used in comparable studies: Peters et al. (2020) with n = 195, Singhal et al. (2025) with n = 300, and Von Zahn et al. (2025) with n = 149 and n = 200 across two studies. Respondents are randomly assigned to one of three experimental conditions to ensure comparability.

The study adheres to ethical research standards: participation is fully anonymous; informed consent is obtained before exposure to any experimental material; respondents may withdraw at any time without consequence; and no personally identifiable information is collected. It is important to acknowledge that convenience sampling limits the generalisability of results to all online shoppers and that the sample is likely biased toward younger, more digitally engaged individuals. This limitation is consistent with comparable studies (Shulner-Tal et al., 2024; Gupta et al., 2026) and will be explicitly addressed in the final thesis's limitations section.

3.6 Data Analysis

Analysis will be conducted in three sequential phases after data cleaning and verification.

Phase 1 — Descriptive Statistics and Balance Checks: Descriptive statistics will summarise participant demographics and key variables. ANOVA balance checks will confirm that random assignment successfully distributed individual differences — prior AI attitude, privacy concern, demographics — evenly across the three experimental groups.

Phase 2 — Experimental Analysis: One-way ANOVA will compare trust scores across the three experimental groups. Post-hoc pairwise comparisons (Tukey HSD) will identify which group differences are statistically significant. Regression analysis will examine how transparency influences trust while controlling for prior AI attitude and privacy concerns. Mediation analysis will test the role of trust as a mediating variable between transparency condition and WTP, using the PROCESS macro (Hayes, 2018). Moderation analysis will test whether the transparency–trust effect is stronger for algorithm-averse consumers, as theorised from Gupta et al. (2026) and Kirshner (2025).

Phase 3 — Conjoint Analysis: Part-worth utility scores will be estimated for each attribute level from the CBC data. Marginal WTP for transparency will be calculated as the ratio of the transparency part-worth utility to the price sensitivity coefficient, producing an estimate of the additional monetary premium consumers would pay for a transparent AI recommendation relative to an opaque one. Subgroup analysis will compare WTP estimates between algorithm-averse and algorithm-appreciative consumers. All statistical analyses will be conducted using Jamovi for experimental testing and Python for conjoint utility estimation. Effect sizes — eta-squared for ANOVA, Cohen's d for pairwise comparisons — will be reported throughout to assess practical significance alongside statistical significance.

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